

DSA Practical No: 01

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Class: C3

INPUT

```
#include <stdio.h>

/* Find length of string */
int length(char str[])
{
    int i = 0;

    while (str[i] != '\0')
        i++;

    return i;
}

/* a. Find substring */
void substring(char str[], int position, int n)
{
    int i;

    printf("Substring: ");

    for (i = 0; i < n && str[position + i] != '\0'; i++)
        printf("%c", str[position + i]);
```

```

    printf("\n");
}

/* b. Check palindrome */
void palindrome(char str[])
{
    int i = 0;
    int j = length(str) - 1;
    int flag = 0;

    while (i < j)
    {
        if (str[i] != str[j])
        {
            flag = 1;
            break;
        }

        i++;
        j--;
    }

    if (flag == 0)
        printf("String is palindrome\n");
    else
        printf("String is not palindrome\n");
}

```

/ c. Compare two strings */*

```
void compare(char str1[], char str2[])
{
    int i = 0;
    int flag = 0;

    while (str1[i] != '\0' || str2[i] != '\0')
    {
        if (str1[i] != str2[i])
        {
            flag = 1;
            break;
        }

        i++;
    }

    if (flag == 0)
        printf("Strings are matching\n");
    else
        printf("Strings are not matching\n");
}
```

/ d. Copy one string into another */*

```
void copy(char str1[], char str2[])
{
    int i = 0;
```

```

while (str1[i] != '\0')
{
    str2[i] = str1[i];
    i++;
}

str2[i] = '\0';
}

```

/ e. Reverse a string */*

```

void reverse(char str[])
{
    int i = 0;
    int j = length(str) - 1;
    char temp;

    while (i < j)
    {
        temp = str[i];
        str[i] = str[j];
        str[j] = temp;

        i++;
        j--;
    }
}

```

```

int main()

```

```

{
    char str1[100], str2[100], str3[100];
    int choice;
    int position, n;
    char ch;

    do
    {
        printf("\n-----\n");
        printf("  STRING OPERATIONS\n");
        printf("-----\n");

        printf("1. Substring\n");
        printf("2. Palindrome\n");
        printf("3. Compare\n");
        printf("4. Copy\n");
        printf("5. Reverse\n");

        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                printf("Enter string: ");
                scanf("%s", str1);

                printf("Enter position: ");

```

```
scanf("%d", &position);
```

```
printf("Enter number of characters: ");
```

```
scanf("%d", &n);
```

```
substring(str1, position, n);
```

```
break;
```

case 2:

```
printf("Enter string: ");
```

```
scanf("%s", str1);
```

```
palindrome(str1);
```

```
break;
```

case 3:

```
printf("Enter first string: ");
```

```
scanf("%s", str1);
```

```
printf("Enter second string: ");
```

```
scanf("%s", str2);
```

```
compare(str1, str2);
```

```
break;
```

case 4:

```
printf("Enter string: ");
```

```
scanf("%s", str1);
```

```
copy(str1, str3);
```

```
printf("Copied string: %s\n", str3);
```

```
break;
```

```
case 5:
```

```
printf("Enter string: ");
```

```
scanf("%s", str1);
```

```
reverse(str1);
```

```
printf("Reversed string: %s\n", str1);
```

```
break;
```

```
default:
```

```
printf("Invalid choice!\n");
```

```
}
```

```
printf("\nDo you want to continue? (y/n): ");
```

```
scanf(" %c", &ch);
```

```
} while (ch == 'y' || ch == 'Y');
```

```
printf("\nProgram ended.\n");
```

```
return 0;
```

```
}
```

OUTPUT:

```
● PS D:\dsa\Experiment-01-String-Operations> .\program.exe
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse

Enter your choice: 1
Enter string: HELLO
Enter position: 2
Enter number of characters: 2
Substring: LL
```

```
● PS D:\dsa\Experiment-01-String-Operations> .\program.exe
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse

Enter your choice: 2
Enter string: MADAM
String is palindrome
```

```
● PS D:\dsa\Experiment-01-String-Operations> .\program.exe
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse

Enter your choice: 3
Enter first string: SAHIL
Enter second string: SAHIL
Strings are matching
```

```
● PS D:\dsa\Experiment-01-String-Operations> .\program.exe
1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse

Enter your choice: 4
Enter string: HELLO
Copied string: HELLO
```

```
● PS D:\dsa\Experiment-01-String-Operations> .\program.exe
```

1. Substring
2. Palindrome
3. Compare
4. Copy
5. Reverse

Enter your choice: 5

Enter string: WORLD

Reversed string: DLROW